

Interface-width and mesh constraints in `comp_eps.py`

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Every equation below is a transcription of `preprocess/comp_eps.py`, which is the single source of truth for ϵ , the mesh, and the derived phase-field coefficients. References: Kaempfer & Plapp, *Phys. Rev. E* **79**, 031502 (2009) [“K&P”] and Moure & Fu, *Cryst. Growth Des.* **24**, 7808 (2024) [“M&F”].

1 Kinetics: the inputs every bound is evaluated against

Thermodynamic state at the run temperature T . Everything below is evaluated once, at the single temperature the mesh is sized for; the solver aborts if the run temperature differs from it by more than 1 °C (`-eps_valid_temp`).

$$\rho_{vs}(T) = \rho_a b \frac{p_{vs}(T)}{p_{atm} - p_{vs}(T)}, \quad \text{saturation vapour density (ASHRAE)} \quad (1)$$

$$D_v(T) = D_{v0} \left(\frac{T}{273.15 \text{ K}} \right)^{1.81}, \quad D_{v0} = 2.178 \times 10^{-5} \text{ m}^2/\text{s} \quad (2)$$

$$d_0(T) = \frac{\gamma (m/\rho_i)}{k_B T} = \frac{\gamma V_m}{RT}, \quad \text{capillary length (K&P Eq. 13)} \quad (3)$$

$$D_i = \frac{\kappa_i}{C_i}, \quad D_a = \frac{\kappa_a}{C_a}, \quad D_{ia}^* = \frac{1}{2} (D_i + D_a), \quad \text{thermal channels} \quad (4)$$

Attachment kinetics. α_c is the model’s only genuinely free parameter. It fixes the Hertz–Knudsen coefficient, and through it every bound in `constraints_eps.tex`:

$$\beta_{HK}(T, \alpha_c) = \frac{1}{\alpha_c} \sqrt{\frac{2\pi m}{k_B T}}, \quad \text{scaled form (K&P } \beta') \quad (5)$$

$$\beta_{sub}(T, \alpha_c) = \frac{\beta_{HK}}{\rho_{vs}(T)/\rho_i} = \frac{\rho_i}{\rho_{vs}(T)} \frac{1}{\alpha_c} \sqrt{\frac{2\pi m}{k_B T}}, \quad \text{unscaled (K&P } \beta_0 \equiv \text{M\&F } \beta_{sub}) \quad (6)$$

The two differ by the factor $\rho_{vs}/\rho_i \sim 10^{-6}$, so mixing them is a six-order-of-magnitude error. Eqs. (9) and (10) below take the scaled β_{HK} ; Eq. (11) takes the unscaled β_{sub} . The solver is passed `-beta_sub0 =  $\beta_{sub}$` and `-d0_sub0 =  $d_0$` and rescales both internally.

Where α_c comes from. Two models are implemented. Libbrecht’s nucleation form makes α_c a function of the local state,

$$\alpha_c(T, \sigma) = \exp \left[-\frac{\sigma_0(T)}{\sigma} \right], \quad \sigma = \frac{|\rho_v - \rho_{vs}(T)|}{\rho_{vs}(T)}, \quad (7)$$

with $\sigma_0(T)$ a log–log interpolant of the Libbrecht (2017) table in $|T|$, held flat beyond both ends, and a hard floor $\alpha_c = 10^{-30}$ whenever $\sigma < \sigma_0/\ln(10^{30})$. Section `constraints_libbrecht.tex` shows why Eq. (7) is not usable here. The fallback is a constant α_c inside the literature band $10^{-3} \leq \alpha_c \leq 10^{-1}$ (Libbrecht 2017; Braun, Fourteau & Löwe 2024).

2 The bounds on ϵ , and the mesh

Width convention. This solver uses Moure & Fu’s well $\frac{1}{2}\phi^2(1-\phi)^2$ with an ϵ^2 gradient term, so the equilibrium profile is logistic,

$$\phi(x) = \frac{1}{1 + e^{-x/\epsilon}} = \frac{1}{2} \left[1 + \tanh\left(\frac{x}{2\epsilon}\right) \right], \quad (8)$$

and ϵ is a *decay length*, not the visible band width: the 5%–95% transition spans $\approx 6\epsilon$ and the 1%–99% transition $\approx 9.2\epsilon$. Kaempfer & Plapp’s ± 1 well has $W = \sqrt{2}\epsilon$; the asymptotic constants used below, $a_1 = 5$ and $a_2 = 0.1581$, are the ones for *this* well.

The four bounds. Three are Kaempfer & Plapp (2009) sharp-interface conditions and carry a common safety factor s (default $s = \frac{1}{2}$); the fourth is geometric and does not.

$$\textbf{B-HEAT:} \quad \epsilon \leq s D_{heat} \beta_{HK} \quad \text{K\&P Eq. (43a/b)} \quad (9)$$

$$\textbf{B-VAPOR:} \quad \epsilon \leq s D_v \beta_{HK} \quad \text{K\&P Eq. (43c)} \quad (10)$$

$$\textbf{B-KINETIC:} \quad \epsilon \leq s \frac{d_0}{\beta_{sub} v_n} \quad \text{K\&P Eq. (45)} \quad (11)$$

$$\textbf{B-CURV:} \quad \epsilon \leq 0.05 R_{ave} \quad \text{geometric, K\&P’s } W \ll 1/\chi \quad (12)$$

and ϵ is taken as the smallest of the four:

$$\epsilon = \min \left\{ s \min (D_{heat}\beta_{HK}, D_v\beta_{HK}, d_0/(\beta_{sub}v_n)), 0.05 R_{ave} \right\}. \quad (13)$$

What each bound is for.

B-HEAT, B-VAPOR control the thin-interface expansion parameter $\delta = a_1 a_2 \epsilon / (D \beta_{HK})$, so on whichever of the two channels binds, $\delta = a_1 a_2 s = 0.79 s$ — that is what the “ $\ll$ ” in K&P Eq. (43) means quantitatively. τ_{sub} already compensates δ to first order, so δ is the size of the correction being applied, not the residual; the residual is $\mathcal{O}(\delta^2)$.

D_{heat} is D_{ia}^* by default — the diffusivity the solver actually assembles and that τ_{sub} compensates against. The single-sided ice channel $D_i = \kappa_i / C_i$ is $\sim 6\times$ tighter and is knowingly violated (enforcing it drove one geometry to $\sim 10^9$ nodes); M&F violate it by $100\times$ in print and say so. It is reported as the diagnostic δ_{ice} instead of enforced.

B-KINETIC keeps the kinetic undercooling $\beta_{sub} v_n$ small against the capillary term d_0/ϵ . It is the *only* bound that involves the front velocity v_n , and the only one that tightens as α_c falls.

B-CURV is deliberately crude. It is one to two decades looser than the binding bound in every case we run, and the smallest true curvature (a fresh neck fillet) starts near-singular and then relaxes, so sizing a mesh from it is both unstable and needlessly expensive.

Mesh. The element size is fixed to ϵ by

$$h = \frac{\epsilon}{\sqrt{2}}, \quad N_d = \left\lceil \frac{L_d}{h} \right\rceil = \left\lceil \frac{\sqrt{2} L_d}{\epsilon} \right\rceil, \quad d \in \{x, y, z\}, \quad (14)$$

i.e. two elements per asymptotic width $W = \sqrt{2}\epsilon$, which is 8.3 elements across $\phi = 0.05$ – 0.95 and 13.0 across $\phi = 0.01$ – 0.99 . Cost therefore scales as ϵ^{-d} : the sizer flags a configuration as *intractable* once $\prod_d N_d > 5 \times 10^7$.

Diagnostics that are reported but never enforced.

$$\delta_X = \frac{a_1 a_2 \epsilon}{D_X \beta_{HK}}, \quad \text{Pe} = \frac{\epsilon v_n}{D_v}, \quad \frac{\epsilon}{R_{ave}}, \quad \frac{\beta_{sub} v_n}{d_0 \chi}, \quad \xi_v \geq 10 \frac{\rho_{vs}}{\rho_i}. \quad (15)$$

ϵ/R_{ave} is noted above 5% and warned above 10%; $\beta_{sub} v_n/(d_0 \chi)$ says whether the Gibbs–Thomson condition is kinetics- or capillarity-dominated; the last is the temporal-scaling condition that keeps the unscaled vapour storage term under $\sim 10\%$ of the source.

3 What is derived from ϵ afterwards

From ϵ to the solver’s coefficients. Once ϵ is fixed, Moure & Fu’s SI Eq. (9) inverts the sharp-interface limit to give the coefficients the solver assembles. With $\rho_{rat} = \rho_{vs}(T)/\rho_i$:

$$d_0 \rho_{rat} = \frac{a_1 \epsilon}{\lambda_{sub}} \implies \lambda_{sub} = \frac{a_1 \epsilon}{d_0 \rho_{rat}}, \quad (16)$$

$$\beta_{HK} = a_1 \left[\frac{\tau_{sub}}{\epsilon \lambda_{sub}} - \frac{a_2 \epsilon}{D_{ia}^*} - \frac{a_2 \epsilon}{D_v} \right] \implies \tau_{sub} = \epsilon \lambda_{sub} \left[\frac{\beta_{HK}}{a_1} + \frac{a_2 \epsilon}{D_{ia}^*} + \frac{a_2 \epsilon}{D_v} \right], \quad (17)$$

and then

$$M_{sub} = \frac{\epsilon}{3 \tau_{sub}}, \quad \alpha_{sub} = \frac{\lambda_{sub}}{\tau_{sub}}, \quad \text{so} \quad \frac{\alpha_{sub}}{M_{sub}} = \frac{3 \lambda_{sub}}{\epsilon} \text{ exactly.} \quad (18)$$

That last identity is why `-mob_sub` and `-alph_sub` are never written into an options file: the solver derives both from $(\epsilon, \beta_{sub}, d_0)$, and overriding either alone breaks the K&P consistency ratio.

Two conditions that must hold for Eq. (17) to mean anything.

$$\tau_{sub} > 0 \quad \text{else the asymptotic expansion is ill-posed;} \quad (19)$$

$$f_{kin} = \frac{\beta_{HK}/a_1}{\frac{\beta_{HK}}{a_1} + \frac{a_2 \epsilon}{D_{ia}^*} + \frac{a_2 \epsilon}{D_v}} \gtrsim 0.5 \quad \text{else } \tau_{sub} \text{ is dominated by thin-interface} \quad (20)$$

corrections rather than by the physical attachment kinetics, and the sharp-interface analogy is being stretched: $f_{kin} < 0.5$ is flagged as marginal, $f_{kin} < 0.1$ as a warning.

Physical constants (identical to `material_properties.c`).

symbol	value	symbol	value
m (H ₂ O molecule)	2.9915×10^{-26} kg	ρ_i	919 kg/m ³
k_B	1.380649×10^{-23} J/K	γ (ice–vapour)	0.109 J/m ²
κ_i	2.29 W/m/K	C_i	1.8×10^6 J/m ³ /K
κ_a	0.02 W/m/K	C_a	1.341×1004 J/m ³ /K
D_{v0}	2.178×10^{-5} m ² /s	a_1, a_2	5, 0.1581

4 Libbrecht (2017) $\sigma_0(T)$ pushed through the chain

The chain. Substituting Eq. (7) into Eq. (6) and then into the kinetic bound Eq. (11) and the mesh rule Eq. (14) gives a single closed expression:

$$N_x = \left\lceil \frac{\sqrt{2} L_x}{\epsilon} \right\rceil, \quad \epsilon \leq s \frac{d_0 \rho_{vs}}{\rho_i} \left(\frac{k_B T}{2\pi m} \right)^{1/2} \frac{1}{v_n} \exp \left[-\frac{\sigma_0(T)}{\sigma} \right] \implies N_x \propto \exp \left[+\frac{\sigma_0(T)}{\sigma} \right]. \quad (21)$$

The mesh is exponential in σ_0/σ . Two facts then decide everything.

1. The supersaturation gap. Libbrecht’s chamber measurements are taken at $\sigma \sim 10^{-2}$ to 10^{-1} . Our problem runs one to two decades lower: the imposed Molaro wall undersaturation is $1 - h = 2.7 \times 10^{-3}$, and the Gibbs–Thomson supersaturation at a neck fillet of radius ρ_f is $\sigma = d_0/\rho_f \approx 4.5 \times 10^{-4}$. Under Eq. (7) that gap is not a two-decade reduction in α_c ; it lands in the exponent.

2. The velocity is measured, not assumed. Eq. (11) bites only if the front moves. We take v_n from Molaro et al. (2019) Fig. 11 itself — the integrated neck-growth rate $v_n = 3.416 \times 10^{-9}$ m/s. So the statement is not “pick a velocity and watch the mesh explode”; it is “the experiment we are replicating moves its interface at this speed, and Libbrecht’s α_c says that is unresolvable.”

The result, at $T = -20^\circ\text{C}$. Molaro grain pair, $L_x \times L_y = 450 \times 225 \mu\text{m}$, $s = 1/2$:

σ	α_c	β_{sub} [s/m]	ϵ [m]	binds	N_x	2D nodes
10^{-1} (Libbrecht)	7.06×10^{-1}	1.12×10^4	4.19×10^{-8}	B-HEAT	1.5×10^4	1.2×10^8
10^{-2} (Libbrecht)	3.07×10^{-2}	2.58×10^5	5.77×10^{-7}	B-KINETIC	1.1×10^3	6.1×10^5
2.7×10^{-3} (ours: wall BC)	2.50×10^{-6}	3.17×10^9	4.69×10^{-11}	B-KINETIC	1.4×10^7	9.2×10^{13}
4.5×10^{-4} (ours: fillet)	10^{-30} (floor)	7.91×10^{33}	1.88×10^{-35}	B-KINETIC	3.4×10^{31}	5.7×10^{62}
<i>For reference:</i> production run $\alpha_c = 0.1$, $\epsilon = 0.118 \mu\text{m}$, $N_x = 5394$, 14.5 M nodes.						
<i>Molecular limit:</i> $a = (m/\rho_i)^{1/3} = 3.19 \text{ \AA}$, so $L_x/a = 1.4 \times 10^6$ — one node per water molecule.						

At our own boundary condition the demand is already 10^{14} nodes with ϵ half an Ångström; at the fillet supersaturation it is 10^{62} nodes with ϵ twenty-five orders of magnitude below a water molecule. This is not an expensive mesh, it is a category error: the continuum phase field has no meaning below a .

Why this is the wrong data, not just an awkward one.

- 1. Libbrecht measured a different experiment.** $\sigma_0(T)$ comes from single ice crystals grown from vapour in a near-vacuum chamber, where the only vapour transport is ballistic and σ is imposed externally and held one to two decades above ours. Our simulations have a continuous vapour field between grains, so σ is *a solution variable*, set by the local curvature and the wall humidity — not a control knob. Extrapolating a nucleation law two decades below where it was calibrated puts the extrapolation entirely inside an exponential.
- 2. The table cannot be fitted anyway.** It is 9 points, it is non-monotonic (a kink at $-6/-7^\circ\text{C}$ that is a digitisation artifact, not physics), and σ_0 spans a factor 27 over -2 to -40°C . Since $\ln \alpha_c = -\sigma_0/\sigma$, no single reference σ can hold α_c inside one decade across that range.
- 3. The escape route is circular.** Feeding v_n back self-consistently from the kinetics, $v_n = d_0/(\beta_{sub} R_{feat})$, makes Eq. (11) collapse to $\epsilon \leq s R_{feat}$ — independent of β_{sub} , and therefore of α_c . The mesh becomes affordable, but only because the constraint has been made insensitive to the quantity under test. The cost simply moves from resolution to runtime: physical t_{final} scales as $1/\alpha_c$.

What Libbrecht’s data *does* establish is the *sign* of the temperature dependence and a plausible band for α_c . That is how it is used: as a consistency check on a constant α_c in $[10^{-3}, 10^{-1}]$, not as a pointwise law.